

EE4C10, Track Specific Assignment: RF

Design of an LNA

(X and Y are the last two digits of your study ID, e.g., so if your study ID is 123456789 (X=8 and Y=9))

Specifications

- Center frequency LNA = $4 + X \cdot (0.2)$ GHz
- BW = 0.7 GHz
- Gain > 8.5 dB
- Current Budget = $10 + Y \cdot (0.2)$ mA
- Supply voltage = 1.8 V
- Noise Figure = 3.0 dB @ 290K (standard noise reference temperature)
- $S_{11} < -20$ dB

Available technology

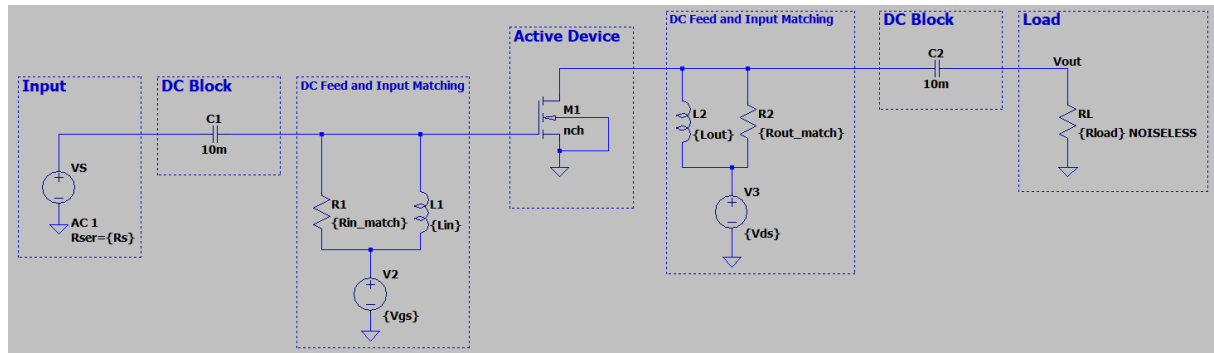
For simplicity reasons, we assume that the capacitances of the active devices have no DC dependency.

In a typical design, you must determine the optimal transistor dimensions and their corresponding biasing conditions yourself. However, we will address this in more detail within the RF track. So, to save you time, we provide you with most of the transistor dimensions below.

- **L = 0.18 μ m**
- **W = to be determined later in the assignment**
- **Number of fingers = 5**
- $Kp = Cox \cdot \mu_n = 2.6 \cdot 10^{-4} A/V^2 \rightarrow$ Kp is the trans-conductance coefficient, which depends on Cox (gate-oxide capacitance per unit area) and the μ_n (electron mobility)
- $C_{ox} = 8.58 \cdot 10^{-3} F/m^2 \rightarrow$ Oxide capacitance per gate area
- $C_{ov} = 3.66 \cdot 10^{-10} F/m \rightarrow$ Overlap capacitance per gate width
- $\lambda = 0.136 \rightarrow$ gate channel-length modulation
- $I_{d,max} = 60 A/m \rightarrow$ maximum current per unit gate width
(beyond this current the device performance drops due to high current effects)

Part I: LNA using shunt resistors

First, we will consider the consequences of using a (too) simplistic design approach toward achieving the required matching condition at the input/output of our “LNA.” For this, we first consider/use the circuit topology below.



As one can observe, a relatively inexperienced RF designer has utilized simple shunt resistors at the input and output of a common-source FET stage to achieve impedance-matched conditions at both the input and output. In addition, inductors have also been added to resonate out the capacitive loading at the input and output.

With the above in mind, try to answer the following questions.

Question 1 – Device size, biasing, and characterization

- To achieve the highest speed for an active device, you must use the highest current density allowed before high current effects set in. For a given current budget, this approach will give you the optimum gate width (W_g) for a FET. For the current budget of 12mA, using the maximum current density indicated for the active devices, what is the optimum W for your FET device?
- Using simple model equations, what is the highest g_m that can be reached for the given current budget? (Tip: Use the long-channel equations provided in the slides)
- For our current budget of 12 mA, choose the appropriate gate bias conditions, which can be done analytically or in simulation by sweeping the gate-source voltage and plotting the drain current vs. V_{gs} , using operating point simulation.
Note: the total gate width in the simulation is the width of one device (W) multiplied by the number of fingers (M). Use 5 fingers, as provided in the template. It is not possible to use one finger, because then the device size is too large.
Now that the FET is correctly biased, check the calculated g_m using LTspice.
- When ignoring the device capacitances, what are the input and output impedances offered by the active device? (Do not include the impedances of the bias networks; they should have hardly any impact if well chosen) Check your calculations in LTSPICE by using a two-port simulation. The simulation set-up is already prepared to run this simulation. Use 'add trace' to plot $Z_{out}(VS)$.
- Calculate C_{gs} and C_{gd} . Take into account the width of the device. Use the equations provided in in slide 29

Question 2 – Matching

Now that we have determined the size of the FET, and calculated its relevant properties, we can try to match the FET to make it function as an LNA.

- A. What should be the value of the resistors (R_1 and R_2) to achieve impedance-matched conditions for the ohmic part at in and output? (ignore the capacitances and inductors for this question, but include the channel modulation)
- B. With your already calculated g_m (part I) and the effective resistors (R_2 and R_L) at the transistor output, calculate now the expected voltage gain of the FET (ignoring R_S).
- C. Calculate the effective input capacitance (C_{in}) of the FET with C_{gs} and C_{gd} , using the previously calculated voltage gain and the Miller approximation.
- D. Using the calculated C_{in} What should be the value of the input inductor (L_1) to achieve conjugate matched conditions at the design frequency (to compensate for the imaginary part due to device capacitances)?

In previous questions, you have determined the component values for the input matching, assuming the output impedance is already at matched impedances. It is also possible to find the capacitance seen from the output; however, in this particular case, the Miller approximation for determining the effective output capacitance does not work due to using an ohmic source with a parallel inductor. Consequently, in this homework assignment, we skip this part (and refer to the RF track for suitable techniques) and use in this assignment $L_{out}=3.9nH$.

Question 3 - Noise

Next, calculate the in-band noise figure F of your LNA analytically at 5.5GHz (slide 41). For this calculation:

- Assume the load (R_L) to be noise-free
- To keep the equations simple, ignore the impact of all reactive elements (remove all capacitors and inductors)
- Since your source impedance is equal to 50Ω (a value different from zero or infinity), your noise sources at the output of the FET should after transformation, result in a noise current source, as well as a noise voltage source at the input of your FET device (for reference, see Razavi page 239). After which, the total noise contributed by the network can be calculated, as well as the noise figure F .
- Since we operate at RF, thermal noise dominates, and the $1/f$ -noise can be neglected.

Question 4 – Simulation

We are going to check our hand calculations using simulations.

- A. First, check the ohmic part of the matching (the shunt resistors). To do so, plot the simulated $Z_{in}(VS)$ and $Z_{out}(VS)$. How do you see whether the ohmic matching is correct?
- B. Check the input and output impedance with the matching inductors. Plot $Z_{in}(VS)$ and $Z_{out}(VS)$. Show that the input and output capacitances are resonated out by the inductors over the intended RF bandwidth.

- C. Plot S_{11} , S_{21} , S_{22} . Do this for the circuit without matching, with ohmic matching only, and with conjugate matching. Note that S-parameters should be plotted in dB. The s-parameters are calculated in LTspice by the two-port simulation. After running the ac-simulation, you can simply add them via 'add trace'.
- D. Plot the noise figure (in dB), again for the circuit without matching, with ohmic matching only, and with conjugate matching. The noise figure equation should be added manually.
- E. Explain very briefly your simulation results and the advantages/disadvantages of adding the resistors and inductors. Indicate what, in your opinion, is the achieved LNA performance using this simplistic "design method." For this purpose, the achieved matching (s_{11}) gain (s_{21}) and the noise figure should also be considered.

Part II: LNA using an inductor in its source connection

The performance of the previously designed LNA was quite bad in terms of gain and noise. Next, we consider the AC equivalent network of an FET with inductive local feedback, as shown below, as an alternative design approach for the LNA.

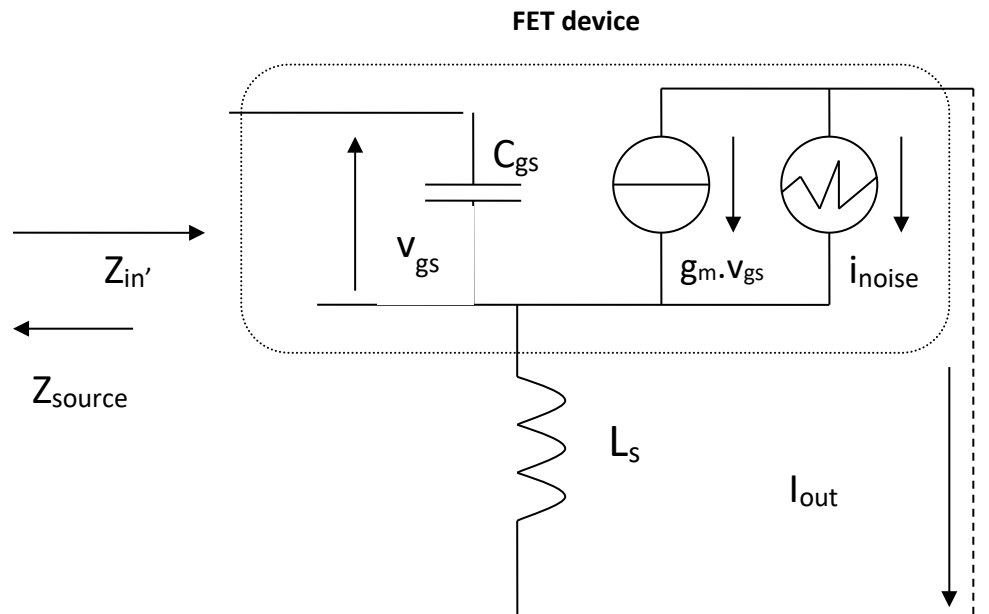


Fig. Strongly simplified AC equivalent schematic of a FET device using local inductive feedback

The source impedance (Z_{source}) is set to 50 ohm. (No simulations are required in answering the following questions).

- Calculate L analytically, such that the real part of $Z_{in'}$ is 50 ohms for your student number specific LNA design frequency and maximum current budget. (For this question, ignore C_{gd} and assume the output to be AC short-circuited.)
- What would you need to do extra (change circuit topology, e.g. by adding circuit elements) to impedance match/cancel also the imaginary part of the $Z_{in'}$, such that the resulting overall input impedance (Z_{in}) of this LNA becomes 50 ohm? (For this question, ignore C_{gd} and still assume the output to be AC short-circuited.)
 - Draw the new LNA input topology, specify its element values!
- When you connect a 50-ohm load at the drain (instead of AC-short circuiting it to ground). What would you expect for the gain and noise performance of this amplifier in comparison with the previous LNA of part 1? Will it be better or worse? Explain why.